



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 13

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Program:

```
def max_subarray_sum(arr):  
    current_sum = arr[0]  
    max_sum = arr[0]  
    for i in range(1, len(arr)):  
        current_sum = max(arr[i], current_sum + arr[i])  
        max_sum = max(max_sum, current_sum)  
    return max_sum  
  
arr = list(map(int, input("Enter array elements: ").split()))  
result = max_subarray_sum(arr)  
print("Maximum subarray sum =", result)
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def max_subarray_sum(arr): 2 current_sum = arr[0] 3 max_sum = arr[0] 4 5 for i in range(1, len(arr)): 6 current_sum = max(arr[i], current_sum + arr[i]) 7 max_sum = max(max_sum, current_sum) 8 9 return max_sum 10 11 arr = list(map(int, input("Enter array elements: ").split())) 12 13 result = max_subarray_sum(arr) 14 15 print("Maximum subarray sum =", result)</pre>			<pre>Enter array elements: -2 -3 4 -1 -2 1 5 -3 Maximum subarray sum = 7</pre>